Olist E-Commerce Platform Analysis Report

1. Problem Understanding

Olist has experienced rapid growth, scaling to over 1 million Reais in monthly revenue between 2016 and 2018. However, this hyper-growth has stressed operational systems. As volume scaled, customer satisfaction trended downward, hitting a platform low of 3.75 stars in early 2018. Leadership requires a comprehensive, data-driven understanding of what is driving customer satisfaction and dissatisfaction across delivery logistics, seller geography, product categories, and payment behavior to guide future scaling strategies.

2. Analytical Approach

The analysis integrated nine operational datasets covering approximately 100,000 orders. The approach moved beyond basic averages to isolate root causes:

- **Delivery Performance:** Calculated absolute delay days by comparing actual delivery timestamps against promised estimates for delivered orders.
- **Geographic Mapping:** Joined customer and seller location data to evaluate the failure rates of in-state versus cross-state shipping pathways.
- **Category Segmentation:** Merged product specifications (weight/dimensions) with review scores to identify friction points in heavy freight.
- **Financial Analysis:** Aggregated payment methods to evaluate the impact of credit card installments on average order value (AOV) and customer expectations.

3. Key Insights

Insight 1: Delivery Delay is the Primary Driver of Dissatisfaction

The data unequivocally shows that missing the estimated delivery date destroys customer satisfaction. Orders that arrive on time or early average a strong **4.29 stars**. However, orders delayed by 1 to 7 days drop to **3.18 stars**, and delays over 14 days plummet to **1.71 stars**. Late delivery is not a secondary factor; it is the root cause of the platform's review score decline.

Insight 2: Cross-State Shipping is Structurally Flawed

Because the majority of sellers are concentrated in the Southeast (São Paulo), orders shipped to the North and Northeast (AL, MA, SE) require complex cross-state logistics. These cross-state routes fail at a significantly higher rate. For example, customers in Alagoas (AL) experience a **24.1% late delivery rate** and pay an average of R$35.87 in freight, directly correlating with lower regional satisfaction.

Insight 3: Heavy Freight Drags Down Platform Averages

Product categories with high average weights suffer from the worst review scores. **Office Furniture** is the lowest-performing significant category on the platform (3.49 stars), weighed down by an average product weight of 11.3 kg and R$40.40 in average freight costs. Conversely, lightweight categories like Books (0.7 kg) achieve the highest scores (4.45 stars).

Insight 4: Installments Drive High-Value Carts

Credit cards dominate the platform, and installment financing is the primary mechanism unlocking high-value purchases. Customers paying in full have an AOV of R$101, while customers utilizing 7 to 10 installments have an AOV of **R$335**. However, these high-value, long-installment purchases carry higher customer expectations, resulting in slightly lower baseline review scores.

4. Business Findings

Dimension	Finding
Marketplace Performance	Hyper-growth in Q4 2017 broke logistics. Revenue scaled, but operational backlog drove average review scores down to 3.75 stars by Q1 2018.
Delivery Satisfaction	A one-week delivery delay guarantees a sub-3.5 star review. Two weeks guarantees a sub-2 star review.
Geographic Patterns	Olist is failing customers in the North and Northeast due to over-reliance on cross-state logistics from Southern sellers.
Category Performance	Heavy, bulky items (Furniture, Bed/Bath/Table) are structurally incompatible with current logistics expectations and require recalibration.
Payment Behavior	Long-term credit card installments are critical for GMV growth, specifically for unlocking expensive, high-friction categories.

5. Actionable Recommendations

1. **De-risk Cross-State Logistics via Localized Supply:** Launch targeted seller acquisition campaigns in the North and Northeast regions to build local inventory hubs, reducing reliance on long-haul transit from São Paulo.
2. **Recalibrate Algorithmic Delivery Estimates for Bulky Freight:** Immediately extend the order_estimated_delivery_date window for heavy categories (e.g., Furniture, Bed/Bath). Setting more conservative delivery expectations will prevent orders from triggering the "late delay" threshold that tanks review scores.
3. **Protect High-Installment Customers:** Implement proactive tracking notifications and priority customer service routing for orders utilizing 4+ credit card installments, as these high-AOV carts are highly sensitive to logistical friction.